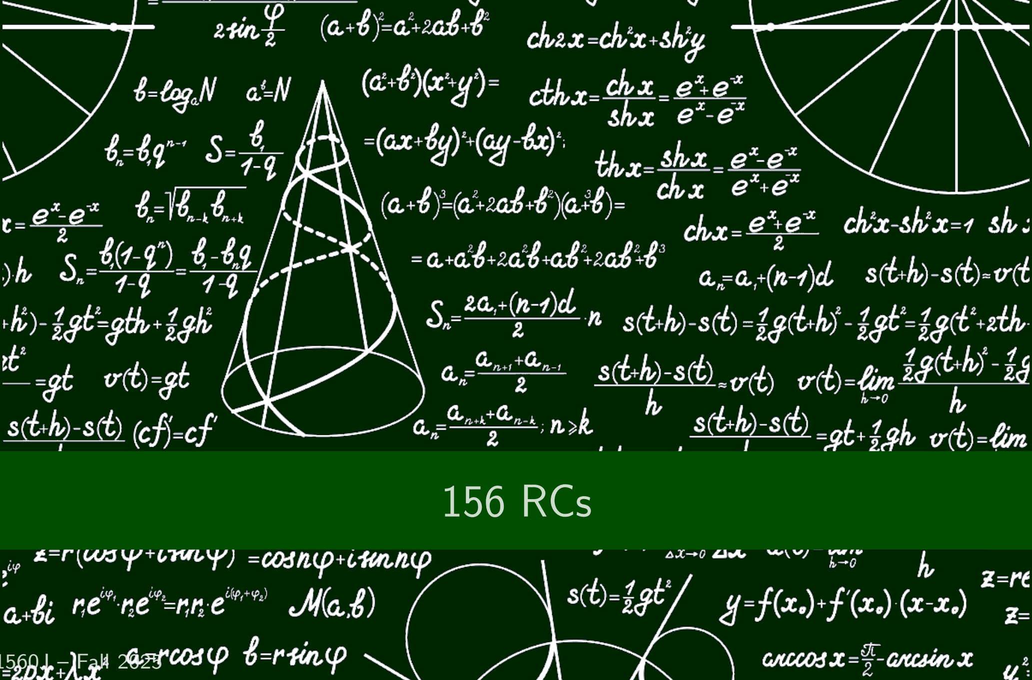


156 RCs

1. Second-Order ODE

Definition

Equations representing the relationship between some unknown functions, the derivative of those functions and the independent variable.

Order: The order of the highest derivative of the unknown function is called the order of the differential equation.

Linear: The highest degree of the unknown function and its derivatives of any order is 1.

Generally, n^th order differential equations have the form of $F(x, y, y', ..., y^{(n)}) = 0$

In VV156, we only need to solve some special types of ODEs.

Step 1: Find the general solution to the corresponding homogeneous ode $y'' + py' + qy = 0$ Solve the characteristic equation: $\lambda^2 + p\lambda + q = 0$

$$\begin{cases} \text{Two different real roots } \lambda_1, \lambda_2 \quad y_h = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x} \\ \text{Two equal real roots } \lambda \quad y_h = C_1 e^{\lambda x} + C_2 x e^{\lambda x} \\ \text{Two different complex roots } \alpha \pm \beta i \quad y_h = C_1 e^{\alpha x} \sin \beta x + C_2 e^{\alpha x} \cos \beta x \end{cases}$$

Step 2: Find one special solution

If $y'' + py' + qy = e^{\alpha x} \cos \beta x$

Let $P(\lambda) = \lambda^2 + p\lambda + q \quad z = \alpha + i\beta$. Take:

1. $y_p(x) = \Re\left(\frac{e^{zx}}{P(z)}\right)$, if $P(z) \neq 0$
2. $y_p(x) = \Re\left(\frac{xe^{zx}}{P'(z)}\right)$, if $P(z) = 0,\ P'(z) \neq 0$
3. $y_p(x) = \Re\left(\frac{x^2 e^{zx}}{P''(z)}\right)$, if $P(z) = 0,\ P'(z) = 0$

Step 2: Find one special solution

If $y'' + py' + qy = e^{\alpha x} \sin \beta x$

Let $P(\lambda) = \lambda^2 + p\lambda + q \quad z = \alpha + i\beta$. Take:

1. $y_p(x) = \Im\left(\frac{e^{zx}}{P(z)}\right)$, if $P(z) \neq 0$
2. $y_p(x) = \Im\left(\frac{xe^{zx}}{P'(z)}\right)$, if $P(z) = 0,\ P'(z) \neq 0$
3. $y_p(x) = \Im\left(\frac{x^2 e^{zx}}{P''(z)}\right)$, if $P(z) = 0,\ P'(z) = 0$

More general method to find particular solution:
In VV156, we only have two types of $f(x)$:

1. $f(x) = e^{\lambda x} P_m(x)$
2. $f(x) = e^{\lambda x}[P_n(x)\cos(\omega x) + P_l(x)\sin(\omega x)]$

For both 1, 2, we first need to check whether $\lambda(\pm\omega i)$ is the root of the characteristic equation $\lambda^2 + p\lambda + q = 0$ or not

$f(x) = e^{\lambda x} P_m(x)$:

1	k
λ is one of the different real roots of $\lambda^2 + p\lambda + q = 0$	1
λ is the identical real roots of $\lambda^2 + p\lambda + q = 0$	2
λ is not the real root of $\lambda^2 + p\lambda + q = 0$	0

$f(x) = e^{\lambda x}[P_n(x) cos(\omega x) + P_l(x) sin(\omega x)]$:

2	k
$\lambda \pm \omega i$ are the complex roots of $\lambda^2 + p\lambda + q = 0$	1
$\lambda \pm \omega i$ are not the complex roots of $\lambda^2 + p\lambda + q = 0$	0

Apply undetermined coefficient method:

- For 1: $y_p = x^k e^{\lambda x} \cdot Q_m(x)$
- For 2: $y_p = x^k e^{\lambda x}[Q_m(x)\cos(\omega x) + R_m(x)\sin(\omega x)] \quad (m = MAX\{n, l\})$

Q_m and R_m are another polynomial of degree m
$\rightarrow$ Calculate y_p' and y_p'' to solve Q_m and R_m

Step 3: Combine y_h and y_p together

If $f(x) = f_1(x) + ... + f_n(x)$, and $f_i(x)$ is the form of 1, 2, we can calculate y_{pi} separately.

Exercise

$$y'' - 4y' + 13y = e^{2x} \cos 3x$$

Answer

1. Step 1: Harmonic Solution
   Characteristic Equation: $\lambda^2 - 4\lambda + 13 = 0$
   Solution: $\lambda_1 = 2 + 3i, \lambda_2 = 2 - 3i$

$$y_h(t) = C_1 e^{2x} \sin 3x + C_2 e^{2x} \cos 3x$$

2. Step 2: Particular Solution
   Let $z = 2 + 3i$
   $z^2 - 4z + 13 = 0,\ 2z - 4 \neq 0$
   Therefore

$$y_p = \Re\left(\frac{xe^{zx}}{2z-4}\right) = \frac{1}{6} xe^{2x} \sin 3x$$

3. Step 3: Final Solution
   $y = y_h + y_p = C_1 e^{2x} \sin 3x + C_2 e^{2x} \cos 3x + \frac{1}{6} xe^{2x} \sin 3x$

2. Parametric Equations & Polar Coordinates

Parametric Equations

Definition

Suppose that x and y are both given as functions of a third variable t (called a parameter) by the equations

$$x = f(t)\ y = g(t)$$

$$x = cost\ \ y = sint\ (0 \leqslant t \leqslant 2\pi)$$

If we plot points, it appears that the curve is a circle. We can confirm this impression by eliminating t. Observe that

$$x^2 + y^2 = sin^2 t + cos^2 t = 1$$

Thus the point (x, y) moves on the unit circle $x^2 + y^2 = 1$. Notice that in this example the parameter t can be interpreted as the angle (in radians). As t increases from 0 to 2π, the point $(x, y) = (cost, sint)$ moves once around the circle in the counterclockwise direction starting from the point $(1, 0)$.

Slope of the Tangent Line with Parametric Curves

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

For the second order derivative, we have:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\frac{dy}{dx} = \frac{d}{dt}\frac{dy}{dx}\frac{dt}{dx} = \frac{\dfrac{d}{dt}\dfrac{dy}{dx}}{\dfrac{dx}{dt}}$$

1. If $y'(t) = 0 \,\&\, x'(t) \neq 0$, we have a horizontal tangent line.
2. If $y'(t) \neq 0 \,\&\, x'(t) = 0$, we have a vertical tangent line.
3. If $y'(t) = x'(t) = 0$, we have to use the definition of slope of tangents. (take limits)

Area

We know that the area under a curve $y = F(x)$ from a to b is $A = \int_a^b F(x)dx$, where $F(x) \geqslant 0$. If the curve is traced out once by the parametric equations $x = f(t)$ and $y = g(t)$, $\alpha \leqslant t \leqslant \beta$, then we can calculate an area formula by using the Substitution Rule for Definite Integrals as follows:

$$A = \int_a^b y dx = \int_\alpha^\beta g(t) f'(t) dt$$

or

$$= \int_\beta^\alpha g(t) f'(t) dt$$

Area

Suppose C is a **simple closed curve** oriented counterclockwise where $x = x(t), y = y(t), \alpha \leq t \leq \beta$ and $x(\alpha) = x(\beta), y(\alpha) = y(\beta)$, then the area enclosed by C is given by

$$\begin{aligned}\text{(unsigned) Area} &= -\int_{\alpha}^{\beta} y(t)x'(t)dt = \int_{\alpha}^{\beta} x(t)y'(t)dt \\ &= \frac{1}{2}\left(-\int_{\alpha}^{\beta} y(t)x'(t)dt + \int_{\alpha}^{\beta} x(t)y'(t)dt\right)\end{aligned}$$

Arc Length

If a curve C is described by the parametric equations $x = f(t)$,$y = g(t)$,$\alpha \leqslant t \leqslant \beta$, where f' and g' are continuous on $[\alpha, \beta]$ and C is traversed exactly once as t increases fromα to β, then the length of C is

$$L = \int_{\alpha}^{\beta} \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2} dt$$

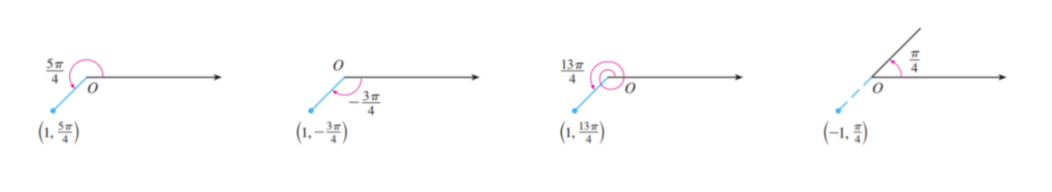
Surface Area

In the same way as for arc length, we can adapt to obtain a formula for surface area. If the curve given by the parametric equations $x = f(t)$,$y = g(t)$,$\alpha \leqslant t \leqslant \beta$ is rotated about the x-axis, where f' and g' are continuous and $g(t) \geqslant 0$, then the area of the resulting surface is given by

$$S = \int_{\alpha}^{\beta} 2\pi|y|\sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2}dt$$

Surface Area

We choose a point in the plane that is called the pole (or origin) and is labeled O. Then we draw a ray (half-line) starting at O called the polar axis. This axis is usually drawn horizontally to the right and corresponds to the positive x-axis in Cartesian coordinates. If P is any other point in the plane, let r be the distance from O to P and let θ be the angle (usually measured in radians) between the polar axis and the line OP as in Figure 1 . Then the point P is represented by the ordered pair (r, θ) and r,θ are called polar coordinates of P. We use the convention that an angle is positive if measured in the counterclockwise direction from the polar axis and negative in the clockwise direction. If P = O, then $r = 0$ and we agree that $(0, \theta)$ represents the pole for any value of θ.



In fact, since a complete counterclockwise rotation is given by an angle 2π, the point represented by polar coordinates (r, θ) is also represented by

$$(r, \theta + 2n\pi) \text{ and } (-r, \theta + (2n+1)\pi)$$

Relationship between Polar Coordinates and Cartesian Coordinates

1. $x = r\cos\theta$
2. $y = r\sin\theta$
3. $r^2 = x^2 + y^2$
4. $\tan\theta = \frac{y}{x}$

Slope of the Tangent Line with Polar Coordinates

$$x = rcos\theta, y = rsin\theta$$

where r can be regarded as a function of θ. Hence we have

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\frac{dr}{d\theta}sin\theta + rcos\theta}{\frac{dr}{d\theta}cos\theta - rsin\theta}$$

Suppose we have the function in polar coordinates:

$$r = f(\theta), a \leqslant \theta \leqslant b \quad r = g(\theta), a \leqslant \theta \leqslant b$$

For the area enclosed by $f(\theta)$, we have:

$$A = \int_a^b \frac{1}{2} f^2(\theta) d\theta$$

For the area enclosed between $f(\theta)$ and $g(\theta)$, we have:

$$A = \int_a^b \frac{1}{2} f^2(\theta) - \frac{1}{2} g^2(\theta) d\theta$$

Suppose we have the function in polar coordinates:

$$r = f(\theta), \alpha \leqslant \theta \leqslant \beta$$

The arc length is given by:

$$L = \int_{\alpha}^{\beta} \sqrt{(\frac{dr}{d\theta})^2 + r^2}\, d\theta$$

Exercise 1

tangents

Find equations of the tangents to the curve $x = 3t^2 + 1, y = 2t^3 + 1$ that pass through the point $(4, 3)$.

Answers

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{6t^2}{6t} = t$$

when pass through $(4, 3)$, we have

$3 - (2t^3 + 1) = t(4 - (3t^2 + 1)) \Rightarrow t = 1$ or $t = -2$

$\Rightarrow y = x - 1$ or $y = -2x + 11$

Exercise 2

1. Find the exact length of the curve
$$x = 1 + 3t^2,\ y = 4 + 2t^3,\ 0 \leqslant t \leqslant 1$$
2. Find the area enclosed by the x-axis and the curve
$$x = 1 + e^t,\ y = t - t^2$$
3. Find the exact area of the surface obtained by rotating the given curve about the x-axis.
$$x = 3t - t^3,\ y = 3t^2,\ 0 \leqslant t \leqslant 1$$

Answers

1. $L = \int_0^1 \sqrt{(\frac{dy}{dt})^2 + (\frac{dx}{dt})^2} dt = \int_0^1 6t\sqrt{1+t^2} dt = \int_1^2 \sqrt{u}(\frac{1}{2} du) = 3 \cdot \frac{3}{2}[u^{\frac{3}{2}}]|_1^2 = 2(2\sqrt{2} - 1)$

2. The curve $x = 1 + e^t$,$y = t - t^2 = t(1-t)$ intersects the x-axis when $y = 0$, that is, when $t = 0$ and $t = 1$. The corresponding values of x are 2 and $1 + e$.
   The shaded area is given by

$$
\begin{aligned}
\int_0^1 (y(t) - 0)x'(t)dt &= \int_0^1 (t - t^2)e^t dt \\
&= \int_0^1 te^t dt - \int_0^1 t^2 \cdot e^t dt \\
&= 3\int_0^1 te^t dt - t^2 e^t|_0^1 \\
&= 3((t-1)e^t)|_0^1 - e \\
&= 3 - e
\end{aligned}
$$

3. $x = 3t - t^3, y = 3t^2, 0 \leqslant t \leqslant 1.$
$(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2 = (3 - 3t^2)^2 + (6t)^2 = (3t^2 + 3)^2$

$$\begin{aligned} S &= \int_0^1 2\pi \cdot 3t^2 \cdot (3t^2 + 3)dt \\ &= 18\pi(\frac{1}{3}t^3 + \frac{1}{5}t^5)|_0^1 \\ &= \frac{48}{5}\pi \end{aligned}$$

Exercise 3

1. Find the slope of the tangent line to the given polar curve at the point specified by the value of θ

$$r = 2 - \sin\theta,\ \theta = \frac{\pi}{3}$$

2. Find the area of the region enclosed by one loop of the curve:

$$r = 2\sin 5\theta$$

Answers

1. $x = r\cos\theta = (2-\sin\theta)\cos\theta, y = r\sin\theta = (2-\sin\theta)\sin\theta$
   $\dfrac{dy}{dx} = \dfrac{2\cos\theta - \sin 2\theta}{-2\sin\theta - \cos 2\theta}$
   when $\theta = \dfrac{\pi}{3}$, $\dfrac{dy}{dx} = \dfrac{2-\sqrt{3}}{1-2\sqrt{3}}$
2. $\sin 5\theta = 0 \Rightarrow 5\theta = n\pi \Rightarrow \theta = \frac{\pi}{5}n$

$$\begin{aligned} A &= \int_0^{\frac{\pi}{5}} \frac{1}{2}(2\sin 5\theta)^2 d\theta \\ &= 2\int_0^{\frac{\pi}{5}} \frac{1}{2}(1-\cos 10\theta)|_0^{\frac{\pi}{5}} \\ &= \frac{\pi}{5} \end{aligned}$$

Reference

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4. Li, Junhao. VV156 RC5.pdf.2022.